

Customer Segmentation Project

Project Report

1. Introduction

Customer Segmentation is a process of dividing customers into different groups based on their behavior, purchasing patterns, demographics, and preferences. This project helps businesses understand customer needs and improve marketing strategies using data analytics.

2. Objective of the Project

The main objective of this project is to segment customers using clustering techniques such as K-Means clustering. It helps identify different customer groups for targeted marketing and improved customer engagement.

3. Tools & Technologies Used

Tool	Purpose
Python (Jupyter Notebook)	Data analysis and clustering
Scikit-Learn	Customer clustering using K-Means
Pandas	Data cleaning and preprocessing
Matplotlib	Visualization of customer segments
CSV/Excel	Customer dataset source

4. Key Features

- Perform clustering using Python (Scikit-Learn) or Power BI.
- Analyze customer purchase patterns and preferences.
- Visualize customer segments and key characteristics.
- Identify target groups for marketing campaigns.

5. Methodology

The project uses customer demographic and purchase data. Data preprocessing is performed to remove missing values and normalize features. K-Means clustering is applied to divide customers into different groups based on similarities.

6. Expected Outcome

This project provides experience in customer analytics, segmentation, and targeted insights. It helps organizations understand customer behavior and improve business strategies.

7. Conclusion

The Customer Segmentation Project demonstrates how machine learning techniques can be used to group customers effectively. Businesses can use these insights to improve customer satisfaction and maximize profits.